Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Advanced Antifouling and Ultra-Low Friction Coating Technologies for Medical Devices			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (√), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1 (**Large Category**)**		**Technology Field 2 (**Large Category**)**	
	Bio Medical		Bio Medical	
	Medium Category	Small Category	Medium Category	Small Category
	Industrial Biotechnology	Functional Biomaterials	Therapeutic Devices & Diagnostic Devices	Clinical Chemistry & Biological Analysis Instruments
Concept of Proposed Technology	Biofouling, friction-induced tissue damage, and foreign body response remain critical limitations for long-term medical device performance and safety. Existing hydrophilic or PEG-based coatings often suffer from insufficient durability, unstable lubrication, or limited resistance to protein adsorption and microbial attachment. This RFP seeks proposals focused on polymer-based surface coating technologies that: • Provide ultra-low friction under hydrated or physiological conditions; • Exhibit strong resistance to protein adsorption and microbial adhesion; • Maintain mechanical integrity and coating stability under long-term physiological and mechanical stress; • Are compatible with scalable manufacturing and diverse medical-device substrates. From the perspective of Johns Hopkins University, a key unmet need in antifouling medical device coatings is the lack of mechanistic understanding linking surface chemistry, molecular organization,			

	and functional performance under hydrated and dynamic physiological conditions. While many coating strategies report low fouling or reduced friction, the underlying structure–property relationships that govern durability, lubrication, and biological interactions remain insufficiently resolved. In particular, Johns Hopkins University brings strong capabilities in spectroscopic surface characterization and microfluidic biointerface platforms, enabling mechanistic investigation of polymer surface behavior under hydrated and dynamic physiological conditions. These approaches allow correlation of surface chemistry and molecular organization with antifouling, lubrication, and durability performance.
Rationale for Joint R&D Project	Successful translation of advanced antifouling coatings requires both fundamental material innovation and industrial-scale process development. • U.S. research institutions contribute expertise in polymer synthesis, surface biointerface science, and advanced material characterization. • Korean startups/SMEs contribute medical-device engineering and commercialization strengths, enabling coating technologies to be developed and practically implemented in real medical-device settings. A Korea–US joint R&D framework enables rapid maturation of coating technologies from laboratory discovery to manufacturing-ready and regulatory-aligned solutions.
Key Words	1. Antifouling Coating, 2. Ultra-low Friction, 3. Biocompatibility, 4. Medical Device, 5. Biofouling Prevention

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (6-7)
R&D Objectives	Proposals should aim to: 1. Develop next-generation antifouling and ultra-low friction coating materials that simultaneously achieve antifouling, low-friction properties; 2. Establish robust and scalable coating processes compatible with medical devices; 3. Demonstrate preclinical-level biocompatibility, durability, and functional performance; 4. Prepare for regulatory submission and global commercialization. Projects are expected to progress from early feasibility (TRL 3–4) toward preclinical validation and manufacturing readiness (TRL 6–7).			
R&D Target Specifications	1. Antifouling and Lubrication Performance • Demonstrate ultra-low friction behavior and significant reduction of protein adsorption and microbial adhesion under physiological conditions. 2. Durability and Stability • Maintain coating integrity and functional performance throughout the intended clinical use period. 3. Manufacturability • Establish reproducible coating processes compatible with scalable and quality-controlled manufacturing across common medical-device. 4. Biocompatibility and Preclinical Readiness • Achieve compliance-ready performance for international biocompatibility and safety standards (e.g., ISO 10993). 5. Commercialization Readiness • Deliver a coating technology suitable for integration into commercial medical devices and future regulatory submission.			
R&D Plan	YEAR 1: Fundamental Polymer Innovation & Early Feasibility (TRL 3–4) • Design and develop novel polymer-based antifouling and ultra-low-friction materials through systematic exploration of chemical composition and structure. • Establish mechanistic understanding of coating-layer formation, surface interactions, and fouling-resistance principles. • Conduct early-stage performance evaluation of friction reduction and antifouling efficacy under controlled in-vitro conditions. • Perform advanced material and surface characterization to define structure–property relationships. • Apply spectroscopic and microfluidic analysis techniques to support mechanistic understanding of coating-layer formation, hydration behavior, and early-stage antifouling and friction performance. • Assess initial feasibility of device compatibility and coating applicability on representative medical-device substrates. • Enable iterative material–process optimization through close Korea–US collaboration. YEAR 2: Pilot-Scale Process Development & Device Integration (TRL 5) • Refine polymer formulations to enhance durability, stability, and functional performance under dynamic and physiologically relevant conditions. • Establish pilot-scale coating processes with reproducible quality, uniformity, and adhesion across representative device geometries. • Validate manufacturability and scalability, including batch-to-batch reproducibility and process robustness.			

	• Integrate coating technology into medical device production workflows aligned with quality management systems. • Prepare coated device prototypes suitable for preclinical-level evaluation. YEAR 3: Preclinical Verification & Commercialization Preparation (TRL 6–7) • Demonstrate preclinical performance, reliability, and durability of antifouling coating technologies under clinically relevant conditions. • Conduct biocompatibility, stability, and long-term antifouling assessments aligned with international standards. • Finalize manufacturing workflows, quality controls, and acceptance criteria for scale-up readiness. • Prepare regulatory-supporting technical documentation and evidence packages. • Establish commercialization and global deployment strategies, including technology transfer and partnership planning.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	• Polymer design and synthesis • Fundamental characterization and mechanistic studies • Early-stage in-vitro performance evaluation • Advanced surface characterization and microfluidic-based evaluation to support mechanistic interpretation of coating performance.
Expected R&R of Korean SME Partner	• Coating process optimization and scale-up • Medical device integration • Preclinical testing, regulatory preparation, and commercialization planning

5. Expected Outcomes/Impact

Expected Outcomes and Impact	• Clinically viable antifouling coating platforms that suppress protein and cell adhesion across diverse medical-device materials. • Reduced infection risk, friction-related injury, and device failure • Strengthened Korea–US collaboration in advanced medical materials • Accelerated global market entry of Korean medical device technologies

6. References

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